TensorFlow.js backend/platforms

TensorFlow.js Node: This repository provides TensorFlow execution in backend JavaScript applications under the Node.js runtime, accelerated by the TensorFlow C binary under the hood. It provides the same API as TensorFlow.js and supports Linux, Mac, as well as the Windows CPU and GPU.

TensorFlow.js WASM: This package adds a WebAssembly backend to TensorFlow.js. It supports models like MobileNet, BodyPix, PoseNet, CocoSSD, AutoML image classification and AutoML object detection.

TensorFlow.js React Native: This package provides the TensorFlow.js platform adapter for React Native. It provides GPU accelerated execution of TensorFlow.js supporting all major modes of tfs usage, which includes the following:

1. Support for both model inference and training
2. GPU support with WebGL via *expo-gl*
3. Support for loading pre-trained models from the Web
4. IOHandlers to support loading models from asyncStorage and models that are compiled into the app bundle

- TensorFlow.js WebGPU: This provides the WebGPU backend

TensorFlow.js models

TensorFlow.js provides lots of pre-trained models, which can be classified as follows.

Images: Under Images, the models are of following types.

- *MobileNet:* Classifies images with labels from the ImageNet database. To install: *npm i @tensoprflow-models/mobilenet.*
- *PoseNet:* Allows for real-time human pose estimation in the browser. To install: *npm i @ tensorflow-models/posenet.*
- *Coco SSD:* An object detection model that aims to localise and identify multiple objects in a single image. To install: *npm i @ tensorflow-models/coco-ssd.*
- *BodyPix:* Enables real-time performance and body part segmentation in the browser using TensorFlow.js. To install: *npm i @ tensorflow-models/body-pix.*
- *DeepLab v3:* This model is for semantic segmentation. To install: *npm i @tensorflow-models/deeplab.*

Audio: Audio has the following model.

- *Speech commands:* This model classifies 1 second audio snippets from the speech commands data set. To install: *npm i @tensorflow-models/speech-commands.*

Text: Under text, the models are of following types.

- *Universal Sentence Encoder:* This encodes text into a 512-dimensional embedding to be used as inputs to natural processing tasks such as sentiment classification and textual similarity. To install: *npm i @tensorflow-models/universal-sentence-encoder.*
- *Text Toxicity:* This ranks the perceived impact a comment might have on a conversation from 'Very toxic' to 'Very healthy'. To install: *npm i @tensorflow-models/toxicity.*

General: General has the following model.

- *KNN Classifier:* This package contains a utility for creating a classifier using the K-Nearest Neighbours algorithm and can be used for transfer learning. To install: *npm i @tensorflow-models/knn-classifier.*

Applications of TensorFlow.js

Gestural interfaces: TensorFlow.js is being used in applications that take gestural inputs with the help of a webcam. Developers are using this library to build applications that translate sign language to speech, enable individuals with limited motor ability to control a Web browser with their face, and perform real-time facial recognition and pose-detection.

Research dissemination: The library has facilitated ML researchers to make their algorithms more accessible to others. For instance, the Magenta.js library, developed by the Magenta team, provides in-browser access to generative music models. Porting to the Web with TensorFlow.js has increased the visibility of their work with their audience, namely, among musicians.

Desktop and production applications: In addition to Web development, JavaScript has been used to develop desktop and production applications. Node Clinic, an open source performance profiling tool, recently integrated a TensorFlow.js model to separate CPU usage spikes caused by the user from those caused by Node.js internals.

The future of TensorFlow.js

Progressive Web apps (PWAs): As PWAs become more popular, we can expect to see more and more integrations with TensorFlow.js and on-device storage. Since TensorFlow.js allows you to save models, you could create a model that trains itself on each user to provide a personalised experience and even works offline.

TensorFlow.js development: With the use of machine learning constantly increasing—and with JavaScript development becoming ever more popular—TensorFlow.js seems like it will only increase in popularity in the near future; so it will probably get new features and updates often. END

References

[1] *https://www.tensorflow.org/js/*
[2] *https://github.com/tensorflow/tfjs*

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